$:= L_0 u + c^+(x) u \Rightarrow Lu - c^+ u = L_0 u \geq 0$

By Case(ii) to $L_0 u$ in $B_2(y) \Rightarrow \frac{\partial u}{\partial \vec{\nu}}(x_0) > 0$

$V(x) = e^{-\alpha|x-y|^2} - e^{-\alpha R^2}$, $\alpha > 0$ to be determined (b) V

$V_{x_i} = -2\alpha(x_i - y_i) e^{-\alpha|x-y|^2}$, $\frac{\partial V}{\partial \vec{\nu}} = V_{x_i}(x_0) \cdot \vec{\nu} = -2\alpha e^{-\alpha|x_0-y|^2} (x_0 - y) \cdot \vec{\nu} < 0$

$V_{x_i x_j} = 4\alpha^2 e^{-\alpha|x-y|^2} \cdot (x_i - y_i)(x_j - y_j) - 2\alpha e^{-|x-y|^2} \delta_{ij}$

$LV = a_{ij}(x) V_{x_i x_j} + b_i(x) V_{x_i} + c(x) V$ $\quad \geq \lambda_0 |x-y|^2 \geq \lambda_0 \rho^2$

$= 4\alpha^2 e^{-\alpha|x-y|^2} \sum_{i,j} (x_i - y_i) a_{ij}(x) (x_j - y_j) - 2\alpha e^{-\alpha|x-y|^2} \sum_{i=1}^n a_{ii}(x)$

$- 2\alpha e^{-\alpha|x-y|^2} \sum b_i(x)(x_i - y_i) + c(x)\left[e^{-\alpha|x-y|^2} - e^{-\alpha R^2}\right]$

$\geq e^{-\alpha|x-y|^2}\left[4\alpha^2 \lambda_0 \rho^2 - 2\alpha n M - 2\alpha |b||x-y| + c(x)\right]$

$\geq e^{-\alpha|x-y|^2}\left[4\alpha^2 \lambda_0 \rho^2 - 2\alpha M(n + \sqrt{n} R) - M\right] > 0$ for α large

2025.10.11 (Strong Maximum Principle SMP)

Condition: • L is strickly elliptic on $\Omega \subseteq \mathbb{R}^n$ bounded

• $\|a_{ij}, b_i, c\|_{L^\infty(\Omega)} \leq M$

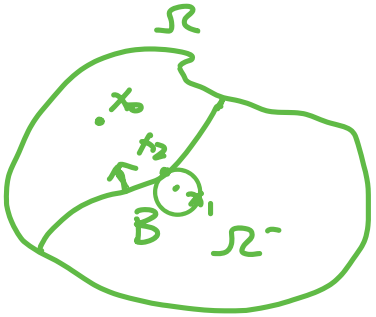
If $u \in C^2(\Omega) \cap C(\bar{\Omega})$ satisfies $Lu \geq 0$ in Ω and $\max_{\bar{\Omega}} u$ is attained at $x_0 \in \Omega$. Then $u(x) \equiv u(x_0)$ in Ω, provided

(1) $c(x) \equiv 0$ or (2) $c(x) \leq 0$ & $u(x_0) \geq 0$ or (3) $u(x_0) = 0$

Pf: Let $\Omega^- = \{x \in \Omega : u(x) < u(x_0)\}$ → open

Goal: show that $\Omega^- = \emptyset$. Suppose $\Omega^- \neq \emptyset$.

Claim: $(\partial\Omega^-) \cap \Omega \neq \emptyset$. Suppose $(\partial\Omega^-) \cap \Omega = \emptyset$.



Let $\Omega_0 = \Omega \setminus \overline{\Omega^-}$ → open. $x_0 \in \Omega_0 \Rightarrow \Omega_0 \neq \emptyset$. Then

$\Omega = \Omega_0 \cup (\overline{\Omega^-} \cap \Omega) = \Omega_0 \cup [\Omega \cap (\Omega^- \cup \partial\Omega^-)]$

$= \Omega_0 \cup (\Omega^- \cup \underbrace{(\Omega \cap \partial\Omega^-)}_{\emptyset}) = \Omega_0 \cup \Omega^-$ & $\Omega_0 \cap \Omega^- = \emptyset$

$\Rightarrow\Leftarrow$ Ω is connected. Hence, $(\partial\Omega^-)\cap\Omega\neq\emptyset$

Let x_1 be a point in Ω^- that is closer to $\partial\Omega^-\cap\Omega$ than to $\partial\Omega$.

Let ball $B\subset\Omega^-$ s.t. x_1 as center. Then $x_2\in\partial B\cap\partial\Omega^-$ $u(x_2)=u(x_0)$

while $u(x)<u(x_0)$, $x\in\overline{B}\setminus\{x_2\}$.

By <u>Hopf Lemma</u>, $\nabla u(x_2)\neq 0\Rightarrow\Leftarrow$ $u(x_2)=u(x_0)$ is global maximum.

Hence, $\Omega^-=\emptyset$. $\Rightarrow u(x)\equiv u(x_0)$ in Ω.

Applications

① Separation of solutions

$\Omega\subseteq\mathbb{R}^n$ bounded, $Lu=a_{ij}(x)u_{x_ix_j}+b_i(x)u_{x_i}$ strickly elliptic

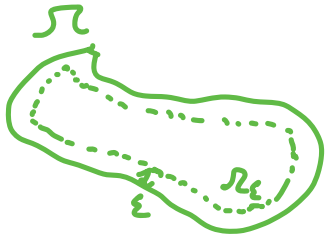
a_{ij}, b_i bounded on Ω. Suppose $u_1, u_2\in C^2(\Omega)$ satisfy $Lu_i=f(x,u_i)$

where f: $\forall M>0$ fixed. $\frac{\partial f}{\partial s}(x,s)$ is uniformly bounded on Ω, $s\in[m,M]$

If $u_1\geq u_2$, $u_1(x_0)=u_2(x_0)$ for some $x_0\in\Omega$. Then $u_1\equiv u_2$ in Ω.



pf: Let $\Omega_\varepsilon=\{x\in\Omega: d(x,\partial\Omega)>\varepsilon\}$ ε small s.t. $x_0\in\Omega_\varepsilon$



only to show $u_1\equiv u_2$ in Ω_ε. Since $\overline{\Omega}_\varepsilon$ is compact.

$u\in C^2(\Omega_\varepsilon)\cap C(\overline{\Omega}_\varepsilon)\Rightarrow\exists M>0$, $|u_i(x)|\leq M$ $i=1,2$. $x\in\overline{\Omega}_\varepsilon$

Let $g(x,t)=f(x, tu_1(x)+(1-t)u_2(x))$, $w=u_1-u_2$

$$Lw=Lu_1-Lu_2=f(x,u_1)-f(x,u_2)=g(x,1)-g(x,0)=\int_0^1\frac{\partial g}{\partial t}\,dt$$

$$=\int_0^1\frac{\partial f}{\partial s}(u_1-u_2)\,dt=w\underbrace{\int_0^1\frac{\partial f}{\partial s}\,dt}:=c(x)w$$

$\Rightarrow Lw-c(x)w=0$, Ω_ε $w\geq 0$ Ω_ε, $w(x_0)=0$

By SMP. $w(x)\equiv w(x_0)=0$ $\Omega_\varepsilon\Rightarrow u_1\equiv u_2$, in Ω_ε. $\varepsilon\to 0^+\Rightarrow u_1\equiv u_2$, in Ω.

② Comparison Principle (C.P.)

- L is strictly elliptic on $\Omega \subseteq \mathbb{R}^n$ bounded.
- $\|a_{ij}, b_i, c\|_{L^\infty(\Omega)} \le M$ with $c \le 0$ in Ω.
- Interior sphere condition satisfied at $x_0 \in \partial\Omega$.

If $u, v \in C^2(\Omega) \cap C(\bar{\Omega})$ s.t. $\begin{cases} Lu \ge Lv & \text{in } \Omega. \\ \frac{\partial u}{\partial \nu} + \beta(x) u \le \frac{\partial v}{\partial \nu} + \beta(x) v & \end{cases}$ $\beta(x) \ge 0, \not\equiv 0$ on $\partial\Omega$

then either $u < v$ in $\bar{\Omega}$ or $u \equiv v$ in $\bar{\Omega}$

Remark: (1) C.P. $\Rightarrow$ Robin B.V.P $\begin{cases} Lu = f & \Omega \\ \frac{\partial u}{\partial \nu} + \beta(x) u = g & \partial\Omega \end{cases}$ has at most one solution

(2) WMP C.P. $\begin{cases} Lu \ge Lv & \Omega \\ u \le v & \partial\Omega \end{cases} \Rightarrow u \le v$ Ω. in case of "$\beta(x) = 0$"

(3) "$\beta \not\equiv 0$". $\begin{cases} \Delta u = \Delta v = 0 & \Omega \\ \frac{\partial u}{\partial \nu} = \frac{\partial v}{\partial \nu} = 0 & \partial\Omega. \end{cases}$ $u = v =$ any constant.

pf of (C.P.): Let $w = u - v$ $\begin{cases} Lw \ge 0 & \text{in } \Omega. \\ \frac{\partial w}{\partial \nu} + \beta(x) w \le 0 & \text{on } \partial\Omega \end{cases}$

Goal: $w < 0$ in $\bar{\Omega}$ or $w \equiv 0$ in $\bar{\Omega}$. Let $M = \max_{\bar{\Omega}} w$

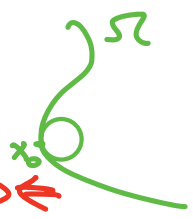
- $M < 0$: $\Rightarrow w < 0$
- $M \ge 0$: If $M = w(x_0)$ $x_0 \in \Omega$ by SMP (ii), $w(x) \equiv M$ $x \in \bar{\Omega}$.

  By boundary condition, $\beta(x) \cdot M \le 0 \Rightarrow \beta(x) \le 0, \beta(x) \ge 0$

  $\Rightarrow M = 0 = w(x)$ in $\bar{\Omega}$.

If $M = w(x_0)$, $x_0 \in \partial\Omega$. x_0 is a strict local max

By Hopf lemma, $\frac{\partial w}{\partial \nu}(x_0) > 0$. By boundary condition, $\Rightarrow \Leftarrow$

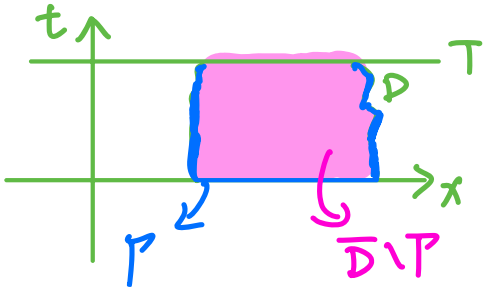


(Classical maximum principle for 2nd order Parabolic Eqs)

Parabolic Eqs: Heat eqs: $u_t - a^2 \Delta u = f(x,t)$ $D \subseteq \mathbb{R}^{n+1}$ $(x,t), x \in \mathbb{R}^n$

bounded, connected

$\bar{D} \cap \{t=0\} \neq \emptyset$, $\bar{D} \cap \{t=T\} \neq \emptyset$.



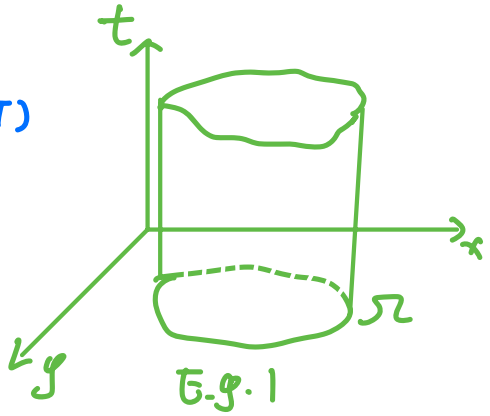
Parabolic boundary of D

$\Gamma = \overline{\partial D \cap \{0 \le t < T\}}$

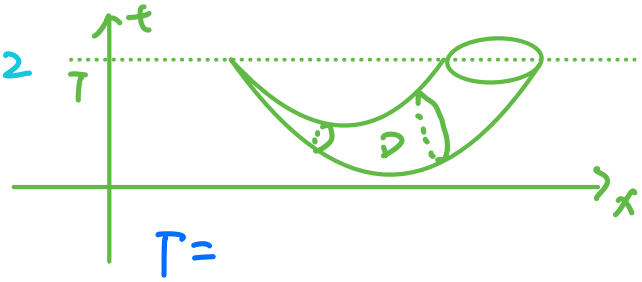
Parabolic interior of D: $\bar{D} \setminus \Gamma$

E.g.1 Ω bounded domain in $\mathbb{R}^n$ $D = \Omega \times (0,T)$

$\Gamma = (\Omega \times \{t=0\}) \cup (\partial\Omega \times \{0 \le t < T\})$

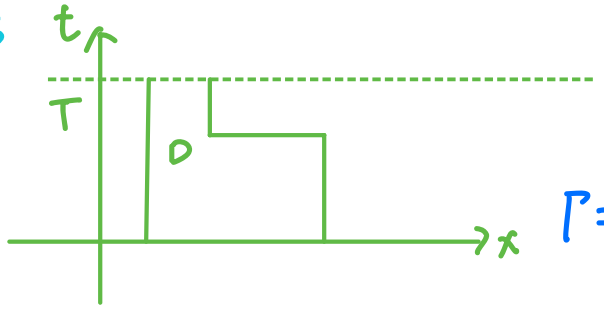


E.g.2



$\Gamma =$

E.g.3



$\Gamma =$